Method for High Spatial Resolution Ventilation Mapping Using Fast-Helical Free Breathing CT

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INTRODUCTION

To reduce the risk of radiation-induced pneumonitis, functional lung avoidance techniques, which seek to consider spatial heterogeneity of lung function during planning, have been proposed. The direct definition of these functional volumes, or ventilation maps, has typically relied on SPECT scans, but previous studies have suggested that ventilation maps can be derived from the planning CT alone. This indirect method of mapping typically relies on breath-hold scans and the 4DCT-modality. FHFBCT allows for data acquisition at each couch position over extended durations, distributing image data across multiple breaths, thus enhancing the likelihood of capturing a representative range of breathing depths and rates when compared to other methods. Utilization of this modality has the possibility of increasing the spatial resolution and accuracy of ventilation mapping.

AIM

To develop a novel method of lung ventilation mapping that employs fast-helical free-breathing CT (FHFBCT) scans.

METHOD

- 194 patients underwent a series of 25-FHFBCT scans with a simultaneous breathing surrogate to track breathing amplitude.
- One scan was chosen to be a reference and the other 24 scans were deformably registered to it using deeds[2,3], then given a follow-up registration using pTVreg[1].
- To measure a voxel's ventilation, we assumed a linear relationship between HU and breathing amplitude and generated a linear fit accordingly, with the breathing amplitude scaled from 0 to 1, corresponding to the 5th and 95th percentile amplitudes respectively.
  - The slope of this linear fit quantified the local ventilation.
  - To reduce the impact of stochastic noise, the HU value for each voxel was taken as the mean of the 3-voxel cube centered at that voxel.
  - Voxels for which the linear fit was poor, as determined by the uncertainty in the slope, had their slopes set to an interpolated value.
    - The interpolation was carried out by seeking the nearest voxel that did not have a poor fit and interpolating to its slope based on the HU difference between the reference voxel and the nearest voxel.

RESULTS

Across all patients:

- Voxels within the lung mask with HU reference values above -50 HU, which were assumed to be soft-tissue, had average slopes of -4±8HUs/breath.
- Voxels with HU values below -50, which were assumed to be parenchymal or gradient regions, had slopes of -42±16HUs/breath.
- The rms of the residuals for all fits was, on average, 11±2-HUs.
- The average lung volume percent that needed to be interpolated due to poor fit quality was 1.25%.

Figure 1 (left): *Histogram of all patients' average ventilation. The global average ventilation is -42 ± 16 HU/breath.*

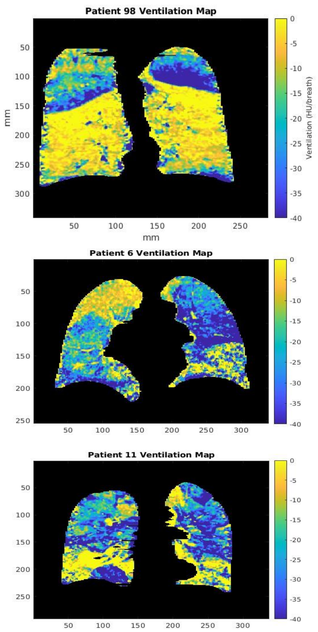


Figure 2 (left): *Displayed are ventilation (units of HU/breath) maps for Patients 98, 6 and 11, chosen for being cases with high-quality DIR and interesting differences in ventilation. In each case we can see areas of low ventilation (more yellow) and regions of high ventilation (more blue). Patient 98 (top), was chosen as the patient had extraordinarily low ventilation due to the lower portions of the lung being almost entirely non-functional. The ventilation map confirms this patient's condition.*

Average ventilation for patients 98, 6, and 11 were -10, -53, and -41 HU/breath respectively.

CONCLUSIONS

- Voxels with HU values within the soft-tissue region had fits with slopes that were approximately zero, corresponding to an expected zero ventilation.
- Voxels within the parenchymal region were shown to have negative slopes with an expected average HU/breath value.

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